

Some Physical Constants

Speed of light	c	$3.00 \times 10^8 \text{ m/s}$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
Coulomb's constant	$1/4\pi\epsilon_0$	$8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
Permittivity constant	ϵ_0	$8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$
Permeability constant	μ_0	$4\pi \times 10^{-7} \text{ N/A}^2$
Planck's constant	h	$6.63 \times 10^{-34} \text{ J}\cdot\text{s}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.673 \times 10^{-27} \text{ kg}$
Neutron mass	m_n	$1.675 \times 10^{-27} \text{ kg}$
Avogadro's number	N_A	6.02×10^{23}

Standard Metric Prefixes (for powers of 10)

Power	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Commonly Used Physical Data

Gravitational field strength $g = \vec{g} $ (near the earth's surface)	$9.80 \text{ N/kg} = 9.80 \text{ m/s}^2$
Mass of the earth M_e	$5.98 \times 10^{24} \text{ kg}$
Radius of the earth R_e	6380 km (equatorial)
Mass of the sun $M_\odot$	$1.99 \times 10^{30} \text{ kg}$
Radius of the sun $R_\odot$	696,000 km
Mass of the moon	$7.36 \times 10^{22} \text{ kg}$
Radius of the moon	1740 km
Distance to the moon	$3.84 \times 10^8 \text{ m}$
Distance to the sun	$1.50 \times 10^{11} \text{ m}$
Density of water [†]	$1000 \text{ kg/m}^3 = 1 \text{ g/cm}^3$
Density of air [†]	1.2 kg/m^3
Absolute zero	$0 \text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}$
Freezing point of water [†]	$273.15 \text{ K} = 0^\circ\text{C} = 32^\circ\text{F}$
Boiling point of water [†]	$373.15 \text{ K} = 100^\circ\text{C} = 212^\circ\text{F}$
Normal atmospheric pressure	101.3 kPa

[†]At normal atmospheric pressure and 20°C .

[‡]At normal atmospheric pressure.

Useful Conversion Factors

1 meter = 1 m = 100 cm = 39.4 in = 3.28 ft
 1 mile = 1 mi = 1609 m = 1.609 km = 5280 ft
 1 inch = 1 in = 2.54 cm
 1 light-year = 1 ly = 9.46 Pm = $0.946 \times 10^{16} \text{ m}$
 1 minute = 1 min = 60 s
 1 hour = 1 h = 60 min = 3600 s
 1 day = 1 d = 24 h = 86.4 ks = 86,400 s
 1 year = 1 y = 365.25 d = 31.6 Ms = $3.16 \times 10^7 \text{ s}$
 1 newton = 1 N = 1 kg·m/s² = 0.225 lb
 1 joule = 1 J = 1 N·m = 1 kg·m²/s² = 0.239 cal
 1 watt = 1 W = 1 J/s
 1 pascal = 1 Pa = 1 N/m² = $1.45 \times 10^{-4} \text{ psi}$
 1 kelvin (temperature difference) = 1 K = $1^\circ\text{C} = 1.8^\circ\text{F}$
 1 radian = 1 rad = $57.3^\circ = 0.1592 \text{ rev}$
 1 revolution = 1 rev = $2\pi \text{ rad} = 360^\circ$
 1 cycle = $2\pi \text{ rad}$
 1 hertz = 1 Hz = 1 cycle/s

1 m/s = 2.24 mi/h = 3.28 ft/s
 1 mi/h = 1.61 km/h = 0.447 m/s = 1.47 ft/s
 1 liter = 1 l = $(10 \text{ cm})^3 = 10^{-3} \text{ m}^3 = 0.0353 \text{ ft}^3$
 1 ft³ = 1728 in³ = 0.0283 m³
 1 gallon = 1 gal = 0.00379 m³ = 3.79 l $\approx$ 3.8 kg H₂O
 Weight of 1-kg object near the earth = 9.8 N = 2.2 lb

1 pound = 1 lb = 4.45 N
 1 calorie = energy needed to raise the temperature of 1 g
 of H₂O by 1 K = 4.186 J
 1 horsepower = 1 hp = 746 W
 1 pound per square inch = 6895 Pa
 1 food calorie = 1 Cal = 1 kcal = 1000 cal = 4186 J
 1 electron volt = 1 eV = $1.602 \times 10^{-19} \text{ J}$

$$T = \left(\frac{1\text{K}}{1^\circ\text{C}}\right)(T_{[\text{C}]} + 273.15^\circ\text{C}) \quad T_{[\text{C}]} = \left(\frac{5^\circ\text{C}}{9^\circ\text{F}}\right)(T_{[\text{F}]} - 32^\circ\text{F})$$

$$T = \left(\frac{5\text{K}}{9^\circ\text{F}}\right)(T_{[\text{F}]} + 459.67^\circ\text{F}) \quad T_{[\text{F}]} = 32^\circ\text{F} + \left(\frac{9^\circ\text{F}}{5^\circ\text{C}}\right)T_{[\text{C}]}$$